

Experiment No: 1

Name of the Experiment: I-V Characteristics of diode.

Objective:

Study the I-V characteristic of diode.

Theory:

A diode is a bi-polar device that behaves as the short circuit when it is in forward bias and as an open circuit when it is in reverse bias condition.

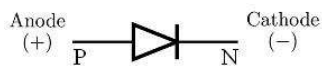


Figure 1.1: Schematic Diagram of Diode.



Figure 1.2: P - N Junction

Diode. There are two types of biasing condition for a diode:

1. When the diode is connected across a voltage source with positive polarity of source connected to P side of diode and negative polarity to N side, then the diode is in forward bias condition.
2. When the diode is connected across a voltage source with positive polarity of source connected to N side of diode and negative polarity to P side, then the diode is in reverse bias condition.

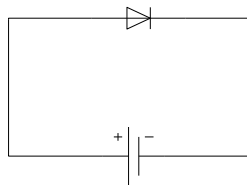


Figure 1.3: Forward Bias connection.

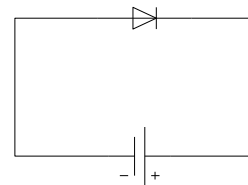


Figure 1.4: Reverse Bias connection.

If the input voltage is varied and the current through the diode corresponding to each voltage are taken, the plot of diode current (I_D) VS diode voltage (V_D) will be follows:

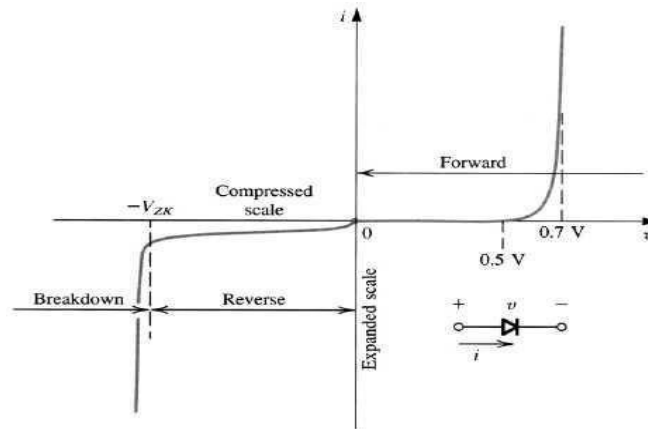


Figure 1.5: I - V Characteristics of Diode.

At the reverse bias condition the amount of current flows through the diode is very small (at microampere range). But if the voltage continuously increases in reverse direction, at a certain value the diode will break down and huge amount of current will flow in reverse direction. This is called breakdown of diode. In laboratory the breakdown will not tested because it will damages the diode permanently.

From the characteristics curve it can be seen that, a particular forward bias voltage (V_T) is required to reach the region of upward swing. This voltage, V_T is called the cut-in voltage or threshold voltage of diode. For Si diode the typical value of threshold voltage is 0.7 volt and for Ge diode is 0.3 volt.

Equipment and Components:

Serial no.	Component Details	Specification	Quantity
1.	p-n junction diode	1N4007	1 piece
2.	Resistor	1K Ω	1 piece
3.	DC power supply		1 unit
4.	Digital Multimeter		1 unit
5.	Chords and wire		as required

Experimental Setup:

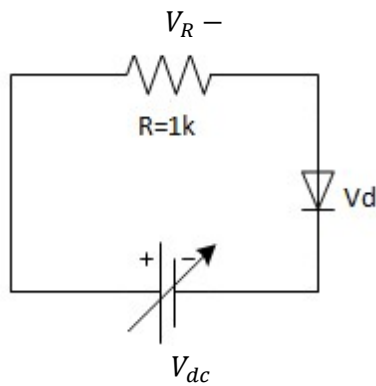


Figure 1.6: Diode forward biased configuration

Procedure:

1. Measure the resistance accurately using DMM.
2. Construct the circuit as shown in Figure 1.6.
3. Vary input voltage V_{dc} with values given in table.
4. Measure V_{dc} , V_d and V_R for each increment of V_{dc} and record data on data table.
5. Obtain maximum value of V_d without exceeding 25V for V_{dc}
6. Calculate the values of I_d using the formula, $I_d = V_R / R$.

NORTH SOUTH UNIVERSITY

DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Data Collection:

Signature of instructor:

Experiment: 1

Performed by Group#

Theoretical value: $R = \text{___ k}\Omega$

Measured value: $R = \text{___ k}\Omega$

V_{dc} (volt)	V_d (volt)	V_R (volt)	$I_d = V_R / R$ (mA)
0.1			
0.3			
0.5			
0.7			
0.9			
1			
2			
3			
4			
6			
8			
10			
12			
14			
16			
18			
20			

Report:

1. Taking readings from the data table, draw I_d VS V_d curve of diode in a graph paper with proper scale [x-axis: 0.2 V per unit, y-axis: any suitable range].
2. What is dynamic and static resistance of a diode?
3. From the graph, find V_d for corresponding values of $I_d = 5$ mA and $I_d = 10$ mA and calculate the static resistance.
4. Considering $V_{dc} = 2$ volt, find the load line (Showing all calculations)
5. Draw the load line in the I_d VS V_d curve of diode and find Q-point.

Objective:

Study the I-V characteristic of diode.

Apparatus List:

Serial No.	Component Details	Specification	Quantity
01	P-N junction diode	1N 4007	1 piece
02	Resistor	1k Ω	1 piece
03	DC power supply		1 Unit
04	Digital Multimeter		1 unit
05	Chords and wire		as required

Theory:

A diode is a two-terminal (Anode-Cathode) semiconductor device that exhibits different behaviours depending on the biasing condition. In forward bias it acts like a short circuit in the circuit. Allowing current flow through it. In reverse bias, it behaves like an open circuit in the circuit and block current flow. This applies for Ideal diodes.

(1) Forward bias:

when the positive terminal of the ^{source} diode is connected to the anode and the negative side is connected to the cathode, the diode conducts current.

(2) Reverse bias:

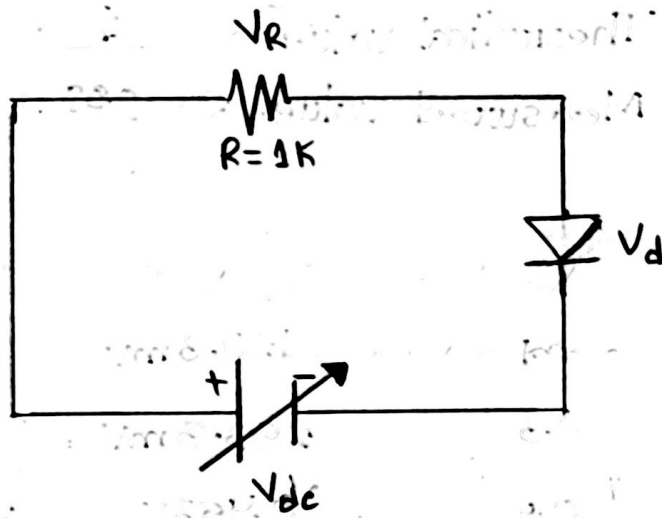
when the positive terminal of the source is connected to the cathode terminal of the diode and negative terminal of the source is connected to the anode terminal of the diode, the diode blocks the current to flow through it.

No bias:

when the diode is not connected to any source and the diode remains neutral.

By varying the input voltage and measuring the corresponding diode current, the relationship between the diode voltage (V_D) and current (I_D) can be determined and plotted.

Circuit Diagram:



Diode forward biased configuration

Data Table:

Theoretical value, $R = 1 \text{ k}\Omega$

Measured value, $R = 0.99 \text{ k}\Omega$

$V_{dc}(\text{Volt})$	$V_d(\text{Volt})$	$V_R(\text{Volt})$	$I_d = \frac{V_R}{R}(\text{mA})$
0.1	120.3 mV	0.0 mV	0.0 mA
0.3	293.3 mV	0.3 mV	0.0 mA
0.5	0.453 V	28.5 mV	0.03 mA
0.7	0.53 V	185.1 mV	0.18 mA
0.9	0.562 V	398.4 mV	0.33 mA
1	0.567 V	0.435 V	0.44 mA
2	0.620 V	1.428 V	1.4 mA
3	0.642 V	2.351 V	2.36 mA
4	0.657 V	3.323 V	3.44 mA
6	0.68 V	5.40 V	5.44 mA
8	0.693 V	7.34 V	7.40 mA
10	0.704 V	9.33 V	9.45 mA
12	0.713 V	11.35 V	11.48 mA
14	0.720 V	13.31 V	13.58 mA
16	0.726 V	15.29 V	16.65 mA
18	0.732 V	17.3 V	17.82 mA
20	0.736 V	19.25 V	20.01 mA

Result Analysis:

The data table shows the general behaviour of a circuit with a diode in forward bias. The table shows that after a certain point the diode doesn't take as much as voltage as before when the source voltage was $(0.4 - 14) V$. The circuit also follows the KVL, $V_{dc} = V_d + V_R$. For the current we applied Ohm's law as after a certain point the diode becomes a short circuit so all the current pass through the diode is equal to the current through the resistor. $I_d = \frac{V_R}{R}$

Questions Answer:

(1) $\Rightarrow$ The graph is shown in the graph paper.

(2) $\Rightarrow$ Dynamic Resistance:

It is the small change in voltage divided by the small change in current in the linear region of the diodes current.

Static Resistance: The ratio of voltage across the diode and current through the diode at a specific point.

(3) $\Rightarrow$ When;

$$I_D = 5 \text{ mA}$$

$$V_D \approx 0.69 \text{ V}$$

$$\text{Static resistance } R = \frac{V_D}{I_D} = 138 \Omega$$

When;

$$I_D = 10 \text{ mA}$$

$$V_D \approx 0.71 \text{ V}$$

$$\text{Static resistance, } R = \frac{V_D}{I_D} = 71 \Omega$$

(4) $\Rightarrow$ We know;

$$V_{DC} = V_D + V_R = V_D + I_D R$$

$$\text{When; } V_{DC} = 2 \text{ V \& } V_D = 0$$

$$I_D = \frac{V_{DC}}{R} = \frac{2}{0.99 \times 10^3} = 2.02 \text{ mA}$$

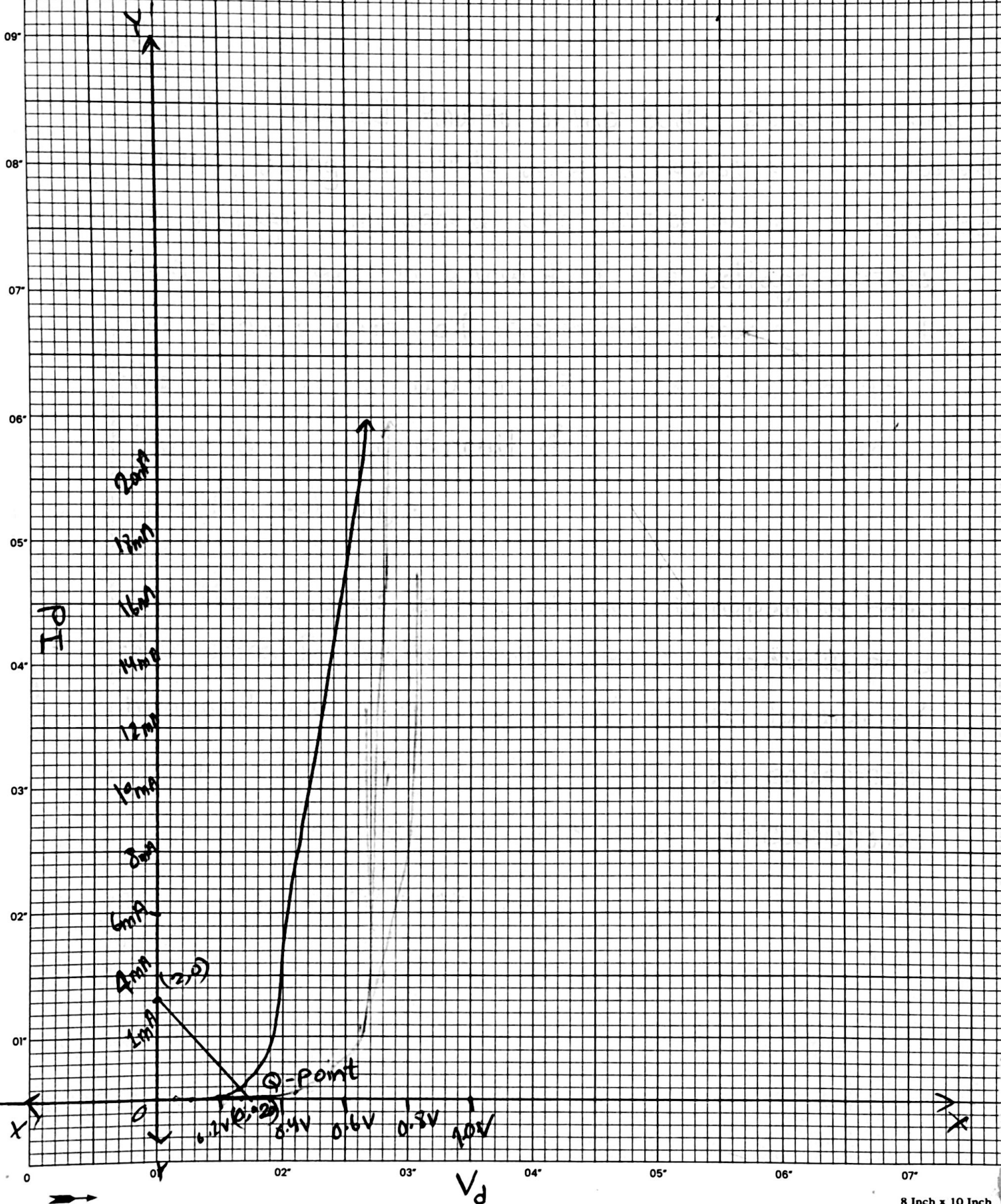
When;

$$I_D R = 0$$

$$V_{DC} = V_D = 2 \text{ V}$$

(5) $\Rightarrow$ from the points $(2, 0)$, $(0, 2.02)$: (V_{DC}, I_D)

The Q-point is $(0.59, 1.2 \text{ mA})$

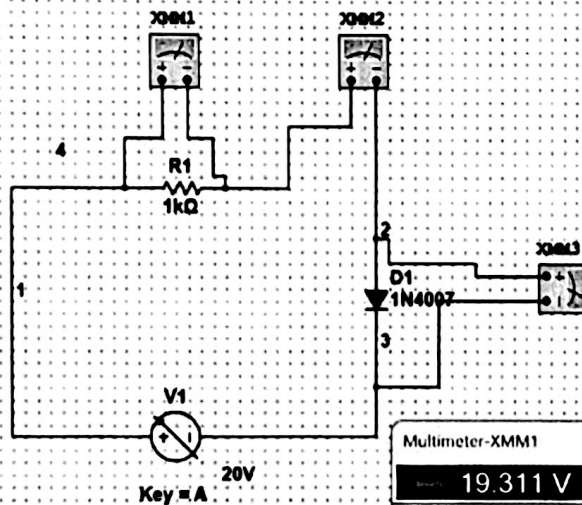


8 Inch x 10 Inch

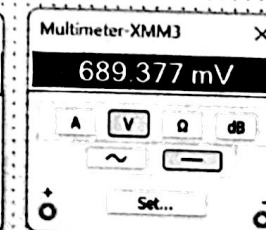
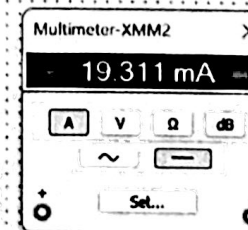
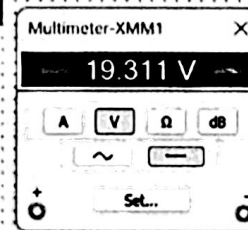
Discussion:

We learned how to setup a dc powered series circuit using a diode and a resistor. We observed the voltage increased behaviour of a diode in forward bias. We got to know that the voltage across the diodes rate of increasing became small when the voltage go up to 0.7V . We also got to know that the circuit follows KVL and Ohm's law (for resistor).

There were some minor discrepancies between theoretical & practical values like resistor value and measured current or voltage. But it was because of static electricity as well as for the equipment. However, the experiment was done with 'cousious' so that the %error is less.



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ID: 2523458643



Data Collection:

Signature of Instructor:

Experiment: 1

Performed by Group# 11

Theoretical value: $R = 1 \text{ k}\Omega$

Measured value: $R = 0.99 \text{ k}\Omega$

[Signature]
16/06/26

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